

Document number: NSI-005-N-E-EH-98-09
Date Revised: 17/Oct/22

Product Name : 2-ETHYLHEXYL ACRYLATE

SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

- PRODUCT NAME : 2-ETHYLHEXYL ACRYLATE

Name of the supplier	:	PT. NIPPON SHOKUBAI INDONESIA
Address	:	Jl. Raya Anyer km. 122, Ciwandan Cilegon 42447 Banten Tel No : 62-254-600660 (Hunting) Fax. No. : 62-254-600657 E-mail Address : sds@shokubai.co.id
Charge section	:	Safety Environment Department
Emergency phone number	:	62-254-600660 (Hunting))
Recommended Use	:	Adhesives and Tackifiers, Paints, Building materials, Acrylic rubbers

2. HAZARD IDENTIFICATION

GHS CLASSIFICATION OF PRODUCT

	<u>CATEGORY</u>
Flammable liquids	: Category 4
Pyrophoric liquids	: Not classified
Acute toxicity (Oral)	: Not classified
Acute toxicity (Dermal)	: Not classified
Skin corrosion/irritation	: Category 2
Serious eye damage /eye irritation	: Not classified
Skin Sensitization	: Category 1B
Germ cell mutagenicity	: Not classified
Carcinogenicity	: Category 2
Specific Target Organ systemic toxicity following single exposure	: Category 3
Specific Target Organ systemic toxicity following repeated exposure	: Category 2
Acute Hazards to the Aquatic environment	: Category 2
Chronic Hazards to Aquatic environment	: Category 2

GHS LABEL ELEMENT**Pictograms****Signal Word : Warning**

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HAZARD STATEMENTS

- Combustible liquid.
- Causes skin irritation.
- May cause allergic skin reaction.
- Suspected of causing cancer
- May cause respiratory irritation
- May cause drowsiness or dizziness
- May cause damage to organs(nasal cavity) through prolonged or repeated exposure
- Toxic to aquatic life.
- Toxic to aquatic life with long lasting effects

PRECAUTIONARY STATEMENTS

[Prevention]

- Obtain special instructions before use
- Do not handle until all safety precautions have been read and understood.
- Keep away from heat, hot surfaces, sparks, open flames and other ignition sources.
No smoking.
- Do not breathe dust/ fume/ gas/ mist/ vapours/ spray.
- Wash hands and face thoroughly after handling.
- Use only outdoors or in a well-ventilated area.
- Contaminated work clothing should not be allowed out of the workplace.
- Avoid release to the environment
- Wear protective glove/protective clothing/eye protection/face protection.

[Response]

- Call a POISON CENTRE/ doctor if you feel unwell.
- Get medical advice/ attention if you feel unwell.
- Take off contaminated clothing and wash before reuse.
- Collect spillage.
- IF ON SKIN : Wash with plenty water.
- IF INHALED: Remove person to fresh air and keep comfortable for breathing.
- IF exposed or concerned: Get medical advice/ attention.
- If skin irritation or rash occurs: Get medical advice/attention.
- In case of fire: use appropriate extinguish media to extinguish.

[Storage]

- Store in a well ventilated place.
- Store locked up.
- Store in a well ventilated place. Keep container tightly closed.

[Disposal]

- Dispose of contents/container in accordance with local/regional/national/international regulation.

Other hazards:

The product can be charged with static electricity.

Even though a polymerization inhibitors is added, polymerization can be initiated by heating, direct sunlight, peroxides, and iron rust. Progress of polymerization can cause temperature increase and accelerated vapor pressure rise, resulting in explosion.

PT. NIPPON SHOKUBAI INDONESIA

KAWASAN INDUSTRI PANCAPURI JL. RAYA ANYER KM 122 CIWANDAN, CILEGON 42447
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3. COMPOSITION / INFORMATION ON INGREDIENTS

Substance or Mixture : Substance

Chemical Name	Content wt. %	CAS Number
2-Ethylhexyl acrylate (Inhibited)	99.8 %	103-11-7

4. FIRST-AID MEASURES

- Inhalation :
Remove person to fresh air and keep comfortable for breathing.
Immediately call a POISON CENTRE/ doctor
- Skin Contact :
Immediately wipe away the attached material with cloths and the like rapidly.
Wash with plenty soap and water.
If skin irritation or rash occurs: Get immediate medical advice/attention.
- Eye Contact :
Flush eyes with water at least 15 minutes. Get medical attention if eye irritation develops or persists.
- Ingestion :
Do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical attention immediately.

5. FIRE-FIGHTING MEASURES

- Suitable Extinguishing media:
Small fires: chemical powder extinguisher, alcohol resistant foam fire extinguishing media, carbon dioxide or water spray.
Large Fire: Alcohol resistant foam fires extinguishing media, water spray or fog.
- Unsuitable extinguishing media:
Direct straight streams (may spread a fire)
- Specific hazards of fire-fighting:
Will be easily ignited by heat, sparks or flames.
Containers may explode when heated. Fire will produce irritating, corrosive and/or toxic gases. May polymerize explosively when heated or involved in a fire. Vapor explosion hazard indoors, outdoors or in sewer.
- Specific extinguishing methods:
Keep upwind of fire.
Use of water spray at a large fire, when another fire extinguishing media may be inefficient.
Fight fire from maximum distance or use unmanned hose holders or monitor nozzles. Cool containers with flooding quantities of water until well after fire is out. For massive fire, use unmanned hose holders or monitor nozzles: if this is impossible, withdraw from area and let fire burn.

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- Special protective actions for fire-fighters:
In case of firefighting, wear positive pressure self-contained breathing apparatus (SCBA) and chemical protective clothing.
Withdraw immediately in case of rising sound from venting safety devices or discoloration of tank.
Structural firefighter's protective clothing will only provide limited protection.

6. ACCIDENTAL RELEASE MEASURES

- Personal Precautions, protective equipment and emergency procedures :
Promptly remove possible ignition sources from the vicinity.
Use personal protection recommended in Section 8. Isolate the hazard area and deny entry to unnecessary and unprotected personnel.
- Environmental Precautions :
Do not flush to sewer or allow entering waterways.
This product has a strong bad smell and irritation. Therefore, take proper measures like informing the people in the neighboring area of the leakage occurrence.
Surround this product with sandbags and earth/sand and cover it with an antistatic sheet to prevent diffusion (of smell).
- Methods and materials for containment and Cleaning Up :
In the case of small amount, hydrolyze it with 5 to 10% sodium hydroxide solution, neutralize, and wash with water. The waste water should be properly treated (incineration, activated sludge process). In the case of large amount, surround it with earth/sand, sandbags and others, cover it with oil-resistant antistatic sheet to prevent generation of vapor, and recover it into proper containers like drums. Then, treat it as in the case of small amount.

7. HANDLING AND STORAGE

- Handling :
 - Technical measures:
This product can be charged with static electricity. Take countermeasures for static electricity removal (grounding, others). Wear antistatic clothes and antistatic shoes to prevent human body electrification.
Use explosion-proof [electrical/ventilating/lighting] equipment.
Use non-sparking tools.
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
Do not handle the containers harshly in such a way as to make them tumbled, dropped, given a shock, or dragged.
 - Local/total ventilation:
Use the ventilation equipment describe in Section 8.
 - Precautions for safe handling:
Not especially.
 - Contact avoidance: Refer to the information in section 10 (Stability and reactivity).
 - Hygiene measures: During handling, be sure to wear proper protective equipment (refer to the section 8).

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- Storage
 - Conditions for safe storage:
 - Keep away from heat.
 - Keep away from sources of ignition – No smoking.
 - Protect from sunlight. Store in a well-ventilated place.
 - When storing this product in a tank, store it as-low-as-possible temperature of 40°C or lower and control to keep the oxygen concentration of the gaseous phase at 5-21%. (Storing at oxygen concentration of lower than 5% can cause polymerization).
 - Safe Packaging material:
 - Store in tightly closed container.
 - Stainless steel or polyethylene container

8. EXPOSURE CONTROL/PERSONAL PROTECTION

- Permissible concentration :

Chemical Name	ACGIH		U.S.-OSHA	
	TLV-TWA	TLV-STEL	PEL-TWA	PEL-STEL
2-Ethylhexyl acrylate (Inhibited)	Not established	Not established	Not established	Not established

- Appropriate Engineering controls:
 - Install eye and body washing facilities near the handling place.
 - Use closed equipment/machinery or use local ventilation equipment.
- Personal Protective Equipment :
 - Respiratory Protection : Gas mask
 - Hand Protection : Organic solvent impermeable protective gloves (antistatic Ones are desirable).
 - Eye/ face Protection : Protective glasses, google, protective face shield
 - Skin and Body protection : Organic solvent impermeable protective clothes, protective shoes (antistatic ones are desirable).

9. PHYSICAL AND CHEMICAL PROPERTIES

[Chemical Product]

- Physical state : Liquid
- Form : Liquid
- Color : Colorless
- Odor : Strong ester odor
- Odor threshold : 0.07 ppm
- Melting point/ freezing point : -90 (°C)
- Boiling Points or initial boiling point and boiling range : 213.5°C
- Flammability : No data
- Lower and upper explosion limit/ flammability limit : 0.65-8.2 (Vol %)
- Flash Point : 88°C Cleveland Open cup
- Auto ignition Temperature : 252°C
- Decomposition temperature : No data

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• pH	: 5.4 (20°C saturated aqueous solution)
• Viscosity	: 1.54 (mPa.s)(25°C)
• Kinematic Viscosity	: No data
• Solubility in Water :	: 25 mg/l (30°C)
• Partition coefficient n-octanol/ water (log value)	: 4.6
• Vapor Pressure	: 23.7 Pa (25°C)
• Density and/or relative density	: 0.8869 (20°C)
• Relative vapour density	: 6.35
• Evaporation rate	: 0.03
• Particle characteristic	: No data
• Other data	: Conductivity: 9.1E-10 S/m Heat accumulation storage Test (UN recommendations): SAPT > 75°C (SAPT: Self- acceleration polymerization temperature)

10. STABILITY AND REACTIVITY :

- Reactivity: Refer to the information in "Possibility of hazardous reactions".
- Chemical Stability :
Even though a polymerization inhibitor is added, heating, direct sunlight, peroxide, and iron rust can cause polymerization.
- Possibility of Hazardous reactions :
When storing at low-oxygen condition of oxygen concentration of lower than 5% at the gaseous phase, polymerization can occur.
- Conditions to avoid :
Heating, contact with incompatible substances, ignition sources.
- Incompatible materials:
Strong oxidizing substance peroxides.
- Hazardous decomposition product :
No available data.

11. TOXICOLOGICAL INFORMATION

[Chemical product]

Acute toxicity, oral	: Not classified LD50: 4430-12800 mg/kg [rat] LD50: 4435 mg/kg [rat]
Acute toxicity, dermal	: Not classified LD50: > 10000 mg/kg [rabbit] LD50: 7522 mg/kg [rabbit]
Acute toxicity, inhalation	: Not classified (Gas) Classification not possible (Vapour) Classification not possible (Dust/Mist)
Skin corrosion/irritation	: Category 2 Severe (rabbit)
Serious Eye damage/irritation	: Not classified Severe (rabbit)

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Sensitization, respiratory	: Classification not possible
Sensitization, skin	: Category 1B
Germ cell mutagenicity	: Not classified
Carcinogenicity	: Category 2 IARC:2B Japan Society for Occupational Health:2B
Reproductive toxicity	: Classification not possible
Specific target organ toxicity (Single exposure)	: Category 3 (anesthetic action, respiratory tract irritation)
Specific target organ toxicity (Repeated exposure)	: Category 2 (nasal cavity)
Aspiration hazard	: Classification not possible

12. ECOLOGICAL INFORMATION

[(Chemical product)]

Eco toxicity	
Acute hazard	: Category 2
Fish	: LC50(96h): 1.81 mg/L [Oncorhynchus mykiss]
Crustacea	: EC50 (48hr) : 1.3 mg/L [Daphnia Magna] EC50 (48hr) : 17.45 mg/L [Daphnia Magna]
Algae or other aquatic plant	: EC50 (96hr) : 47 mg/L [Desmodesmus] EC50 (72hr) : 44 mg/L [Desmodesmus]
Chronic hazard	: Category 2
Fish	: No data
Crustacea	: NOEC(21day): 0.136 mg/L [Daphnia magna]
Algae or other aquatic plants	: No data
Persistence and degradability	: Rapidly biodegradable
Bioaccumulative potential	: Bioaccumulation
Mobility in soil	: No data
Hazardous to the ozone layer	: Classification not possible

13. DISPOSAL CONSIDERATION

Disposal method:

When waste material and waste water are to be treated, incinerate them or collect them into specified containers and entrust the disposal to a licensed industrial waste disposal company.

Do not use the used containers for other purposes like filling other substances. After treating the content according to the above description, the empty containers should be reused, recycled, or disposed through suitably qualified or licensed contractor, in accordance with national/local regulations.

The waste water can be treated also by an activated sludge method.

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14. TRANSPORT INFORMATION

- ICAO/IATA
 - UN classification : Class9
 - UN number : 3082
 - Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE,
LIQUID, N.O.S., STABILIZED
(2-Ethylhexyl acrylate)
 - Packing group : III
- IMDG
 - UN classification : Class9
 - UN number : 3082
 - Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE,
LIQUID, N.O.S., STABILIZED
(2-Ethylhexyl acrylate)
 - Packing group : III
- Marine pollutant : Yes
- Transport regulations :
Transport this product in accordance with local/ regional/ national/ international regulations.
- Special precaution:
At the time of discharge from a tank truck, be sure to ground to prevent static electricity charge and to completely empty the remaining liquid in the pipes.
Load the containers in such a way as not to fall down, tumble, or being damaged.
Cover the loaded cargo to prevent direct sunlight.
(Reference information) Heat accumulation storage test (UN recommendations)
SAPT > 75°C (SAPT : Self-acceleration polymerization temperature)

15. REGULATORY INFORMATION

- Inventory
 - Japan (ENCS) : Listed
 - Korea (KECL) : Listed
 - Australia (AIIC) : Listed
 - Canada (DSL) : Listed
 - China (IECSC) : Listed
 - EU (EINECS) : Listed
 - New Zealand (NZIoC) : Listed
 - USA (TSCA) : Listed
 - Philippine (PICCS) : Listed
 - Taiwan (TCSI) : Listed
- Restriction at export (Indonesia) : Not Applicable

16. OTHER INFORMATION

- Date prepared : 01/Sept/1998
- Date revised : 17/Oct/22
- Date review : 17/Oct/27
- Revise No : 09

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- **Disclaimer:**
 - The described information has been prepared based on the latest materials, information, and other data available at the time of preparation: however, it does not intend to guarantee anything like content, physical/chemical properties, and danger/hazard. Also, the precautions mentioned above are for ordinary handling: therefor, for special handling, it is obligation of each user of the product to take proper measures suitable for each handling condition. Take enough care for such matters.
 - This SDS is based on JIS Z7252: 2019 and JIS Z7253: 2019. If "Not classified" is provided in the GHS classification in Section2, it means that sufficient information is available to classify the product and sufficient evidence is found to determine that it is not applicable to any hazard categories defined by JIS Z7252. And another case is that it based on comprehensive judgment by experts.
 - Regulatory information with regard to this substance in your country or region should be examined by your own responsibility